Completing the square

1 Halve the coefficient of x and then square it.

2

a No

b 9

3 $\frac{9}{4}$

4

a 25

b 1

c 8

d $\frac{1}{4}$

e $\left(\frac{m}{2}\right)^2$

f $\frac{4}{25}$

5

a $x^2+4x+4=(x+2)^2$

b $x^2-5x+\frac{25}{4}=\left(x-\frac{5}{2}\right)^2$

c $x^2-\frac{7}{4}x+\frac{49}{64}=(x-\frac{7}{8})^2$

6

a $(x+6)^2-36$

b $(x-6)^2-36$

c $(x+1)^2+1$

d $(x-7)^2+7$

e $(x-3)^2-4$

f $\left(x+\frac{3}{2}\right)^2+\frac{15}{4}$

g $\left(x-\frac{7}{2}\right)^2+\frac{7}{4}$

7

a $y=(x+53)(x+3)$

b $y=(x-3)(x-21)$

8

a $y=\left(x-3+\sqrt{2}\right)\left(x-3-\sqrt{2}\right)$

b $y=\left(x+4+\sqrt{2}\right)\left(x+4-\sqrt{2}\right)$

Non-monic completing the square

9 $4\left(\left(x-\frac{9}{2}\right)^2-\frac{1}{2}\right)$

10

a $3\left(x+\frac{3}{2}\right)^2+\frac{5}{4}$

b $4\left(x-\frac{11}{8}\right)^2-\frac{9}{16}$

11 $y = (2x + 23)(x + 1)$

Solving equations

12

a No
b Yes
c No
d No
e Yes

13

a $x = -4, -8$
b $x = \frac{3}{2}, \frac{1}{2}$
c $x = 5, 1$
d $x = \pm 2\sqrt{5} - 2$
e $x = -5 \pm \sqrt{17}$
f $x = -9 \pm \frac{\sqrt{30}}{2}$

14

a $x = -2, -16$
b $x = 4, 2$
c $x = 8, 1$
d $x = 8, -2$
e $x = -1, -\frac{7}{4}$
f $x = 1 \pm \sqrt{33}$

15

a $x = -12 + \sqrt{139}, -12 - \sqrt{139}$
b $x = \frac{-11 \pm \sqrt{101}}{2}$
c $x = \pm 2\sqrt{3} - 4$
d $x = \frac{7}{2} \pm \frac{\sqrt{17}}{2}$
e $x = -\frac{1}{6} \pm \frac{\sqrt{109}}{6}$
f $x = \frac{-11}{2} \pm \frac{\sqrt{2965}}{10}$

Applications

16 $x = 3, -7$

17 $k = \pm 8$

18 $x = 12$